CLINICAL · EAR

External ear tumours and bony growths

The pinna faces the sun every day of a person's life, and most cancers of the ear start there. Any lesion of the pinna that does not heal deserves a biopsy. In the canal, a hard lump is often just bone, an osteoma or exostoses, while a painful, bleeding canal may be carcinoma.

AT A GLANCE

- Most ear cancers start on the sun-exposed pinna.
- Biopsy any non-healing lesion.
- Canal SCC is staged by the Pittsburgh system.
- Bony growths need treatment only if they obstruct.

1 CHAPTER ONE Overview

Tumours of the pinna and canal.

1.1 Classification

EXTERNAL EAR TUMOURS

- PINNA, BENIGN **Cysts, keloid** haemangioma, papilloma, cutaneous horn, keratoacanthoma, neurofibroma
- PINNA, MALIGNANT **Basal cell carcinoma, squamous cell carcinoma** melanoma
- CANAL, BENIGN **Osteoma, exostoses** ceruminous adenoma, papilloma
- CANAL, MALIGNANT **Squamous cell carcinoma** basal cell carcinoma, adenoid cystic and ceruminous carcinoma, melanoma

1.2 Where ear cancers arise

About 85% on the pinna, 10% in the canal and 5% in the middle ear.

CHAPTER TWO

2 Benign lesions of the pinna

Lumps that are not cancer.

2.1 Keloid

Overgrowth of scar after piercing, more common in darker skin. Excision alone recurs; combine with steroid injection, pressure clips or, rarely, radiotherapy.



FIGURE 1 Keloid after ear piercing.
Dhingra, fig. 16.1

WHERE THE BOOKS DIFFER · RADIOTHERAPY FOR KELOID

Dhingra	Lee
Postoperative radiation is an option.	Radiation is rarely used.

2.2 Haemangioma and vascular malformations

A *strawberry haemangioma* grows in the first year and regresses by about five years; a *port-wine stain* does not. Venolymphatic malformations may need excision.

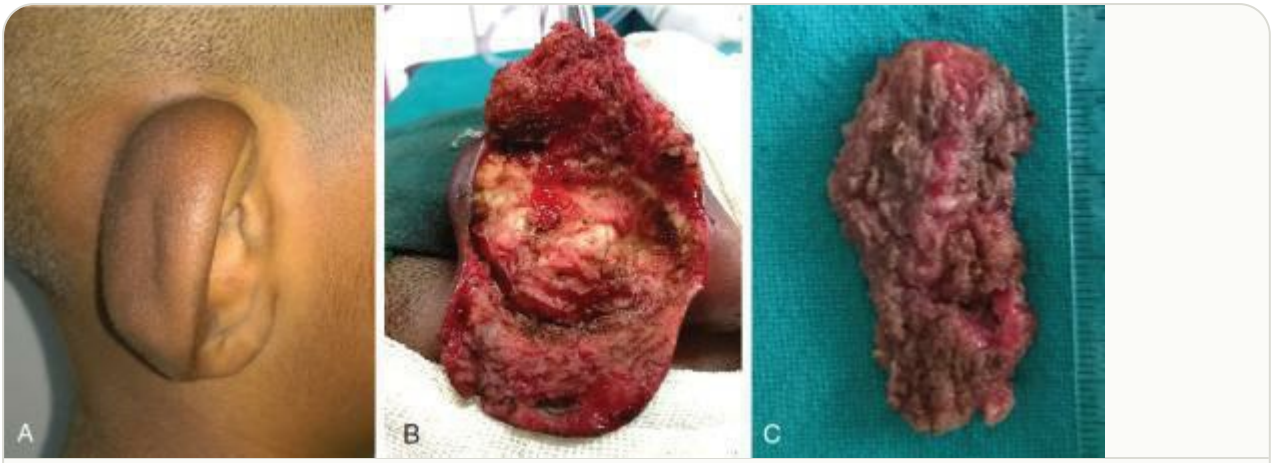


FIGURE 2 A venolymphatic malformation of the pinna in a child: before, during operation, and the excised specimen.

Dhingra, fig. 16.2

2.3 Keratoacanthoma

A rapidly growing crater with a keratin plug that may regress on its own; it looks malignant and is usually excised.

2.4 Others

Sebaceous and dermoid cysts, papillomas, cutaneous horns and neurofibromas.

3 CHAPTER THREE Malignant tumours of the pinna

Sun-exposed skin.

3.1 Risk factors

Sun exposure, fair skin, male sex, older age and immunosuppression.

3.2 Squamous cell carcinoma

An ulcer, usually on the helix, with raised, everted edges. It spreads to the preauricular, postauricular, parotid and upper cervical nodes.



FIGURE 3 Squamous cell carcinoma of the pinna.
Dhingra, fig. 16.3

3.3 Basal cell carcinoma

A nodule with a pearly, rolled, beaded edge and central crusting, usually behind the ear or on the helix; it rarely spreads to nodes but invades locally.

WHERE THE BOOKS DIFFER · THE COMMONEST EAR MALIGNANCY

Dhingra	Lee
Emphasises squamous carcinoma on the helix.	Basal cell carcinoma is the commonest, at about 45%.

3.4 Melanoma

A pigmented, irregular lesion; spreads early to the parotid and neck nodes.

3.5 Treatment

- **Squamous carcinoma:** excision with about 1 cm margins for small lesions; larger lesions, those within 1 cm of the canal, or with nodes need partial or total amputation

of the pinna with parotidectomy and neck dissection.

- **Basal cell carcinoma:** radiotherapy for superficial lesions not involving cartilage; excision (Mohs surgery where available) for deeper lesions.
- **Melanoma:** wedge excision if small on the helix; wider resection with parotid and neck surgery as staged.

OPERATIVE GUIDES IN THEATRE 3 operations

Osteomas and Exostoses of the External Auditory Canal
Osteoma and exostoses

Resection of the Temporal Bone
Temporal bone resection

Management of Skin Cancer of the Head and Neck
Skin cancer of the head and neck

CHAPTER FOUR

4 Bony growths of the canal

Often just bone.

OSTEOMA OR EXOSTOSES?

	OSTEOMA	EXOSTOSES
Number	Single	Multiple
Sides	One	Usually both
Shape	Smooth, pedunculated	Sessile, broad-based
Site	Outer canal, at the bony-cartilaginous junction	Deep canal, near the drum
Cause	True benign tumour	Reaction to cold water (swimmers, surfers)



FIGURE 4 An osteoma arising from the anterior wall of the right ear canal (arrow).
Dhingra, fig. 16.4

WHERE THE BOOKS DIFFER · SITE OF EXOSTOSES

Dhingra	Lee
Deep in the canal, near the drum.	On the anterior and posterior canal walls.

4.1 Treatment

None if small. When they block the canal, trap wax or cause recurrent otitis externa: an osteoma can be removed by fracturing its pedicle or drilling; exostoses are drilled away through the canal or a postauricular approach.

> **PEARL**
> Use a drill, not a gouge and hammer, for exostoses: the facial nerve lies close to the posterior canal wall.

5 CHAPTER FIVE Tumours of the canal

Glands and skin.

5.1 Ceruminous gland tumours

A firm, skin-covered mass in the outer canal. Malignant ones (adenoid cystic carcinoma, ceruminous adenocarcinoma) outnumber benign ones about two to one. Treated by wide excision, with radiotherapy for malignant tumours.

5.2 Squamous cell carcinoma of the canal

Often follows long-standing chronic discharge. It presents with bloody discharge, pain and granulation that will not heal; facial palsy means advanced disease.

5.2.1 Staging (Pittsburgh)

- **T1:** confined to the canal, without bone erosion.
- **T2:** limited bone erosion, or soft tissue extension under 0.5 cm.
- **T3:** full-thickness bone erosion with soft tissue under 0.5 cm, or middle ear or mastoid involvement.
- **T4:** cochlea, petrous apex, medial wall, carotid canal, jugular foramen, dura, soft tissue over 0.5 cm, or facial palsy.

5.2.2 Treatment

Sleeve or lateral temporal bone resection, with parotidectomy and neck dissection as needed, and postoperative radiotherapy.

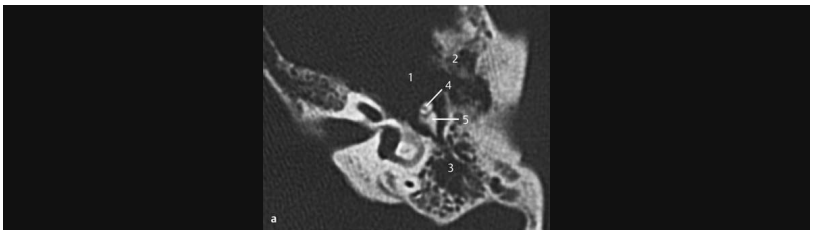
6 CHAPTER SIX Assessment

Biopsy and image.

- Biopsy any non-healing lesion of the pinna or canal.
- CT and MRI for deep or canal tumours.

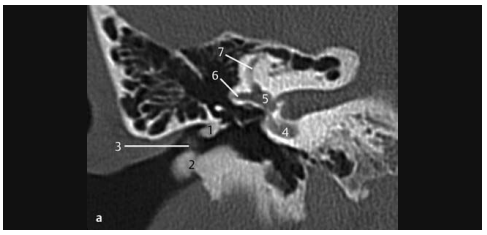
IMAGING IN THE APP 2 cases

Canal carcinoma and exostoses.



Malignancy of the External Auditory Canal Squamous Cell Carcinoma

Tumours · 2 images



Exostoses of the External Auditory Canal

Bone disease · 2 images

- Examine the anterior and posterior auricular, parotid and upper cervical nodes.

6.1 Differential diagnosis

Chondrodermatitis nodularis helicis, perichondritis, necrotising otitis externa and granulation from chronic otitis.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Squamous versus basal cell carcinoma** of the pinna, and their treatment.
- **Osteoma versus exostoses.**
- **Pittsburgh staging** of canal carcinoma.
- **Biopsy any non-healing lesion.**

7.2 Common pitfalls

COMMON PITFALLS

- Treating a non-healing helical ulcer with creams for months.

- Using a gouge on exostoses.
- Missing canal carcinoma in a patient with chronic discharge and new pain.

Written for this app from Oxford Textbook of Otolaryngology ch. 13, 18, Oxford Specialist Handbook ch. 15, Dhingra ch. 16 and K. J. Lee ch. 23. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

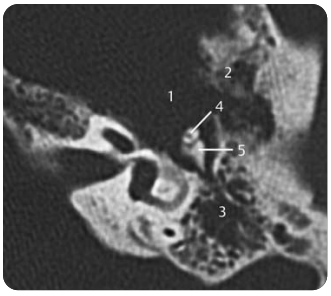
On imaging

The imaging of this topic: key findings, differentials and representative scans.

EXTERNAL EAR & TYMPANIC MEMBRANE · TUMOURS

Malignancy of the External Auditory Canal Squamous Cell Carcinoma

- Extension of bony destruction and features suggestive of infiltrative growth are associated with high morbidity, such as dysfunction of the facial nerve and inner ear.
- Adherence to vital structures, such as the internal carotid artery, may preclude surgery.
- Secondary complications, i.e., mastoiditis or otitis media, as a result of stasis of secretions due to obstruction of the eustachian tube. After radiotherapy, necrotizing osteomyelitis and mastoiditis may occur.

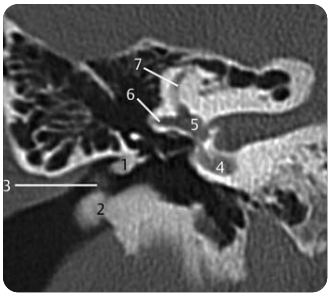


CT, axial. Cranial slice at the level of the horizontal canal, showing destruction of the wall of anterior epitympanum (1), irregular invasion of…

EXTERNAL EAR & TYMPANIC MEMBRANE · BONE DISEASE

Exostoses of the External Auditory Canal

- In patients with aural discharge, chronic otitis may be the result of infected epithelial stasis.
- In cases with a narrowed orifice and meatus, a meatoplasty might be considered for better aeration and cleaning options. Furthermore, CT might be reassuring, showing a normal aerated middle ear.



CT, axial. Small exostoses are visualized in the roof of the external auditory canal near the annulus (1), and a larger one laterally on the floor …